

Mini Lesson Sample Script

1. Multiply a positive number by a negative number using a number line to determine the sign of the product.

Evaluate $4 \times (-3)$ using the number line.



Answer: -12

T: We're going to evaluate $4 \times (-3)$. We can read this as 4 groups of -3. Let's use the number line. I'll start at 0 and jump -3 once. Where do I land?

S: At -3.

T: That's one group of -3. Now you do the next three jumps. Each one moves -3.

S: (Jumps to -6, then -9, then -12.) I landed at -12.

T: So what is $4 \times (-3)$?

S: -12.

T: A positive number times a negative number gives a negative product. You knew this from earlier work, and the number line shows why: four jumps of -3 move us into the negatives.

2. Use the distributive property to discover that the product of two negative numbers is positive.

Evaluate $-4 \times (5 + (-2))$ two ways.

First, evaluate inside the parentheses, then multiply.

Then, use the distributive property to rewrite the expression as the sum of two products, and evaluate.

Compare the two results to determine the value of $-4 \times (-2)$.

Answer: First way: $-4 \times 3 = -12$; Second way: $-20 + 8 = -12$, so $-4 \times (-2) = 8$

T: Let's evaluate $-4 \times (5 + (-2))$ two ways. Start with what's inside the parentheses. What is $5 + (-2)$?

S: 3.

T: So now we have -4×3 . What's that?

S: -12. A negative times a positive is negative.

T: Good. So the answer to the whole expression is -12. Now let's solve it a different way, using the distributive property. How can you rewrite $-4 \times (5 + (-2))$?

S: $(-4 \times 5) + (-4 \times (-2))$.

T: Evaluate the first part.

S: -4×5 is -20.

T: Now think carefully. We know the whole expression equals -12. So -20 plus the product of $-4 \times (-2)$ has to equal -12. What does $-4 \times (-2)$ need to be?

S: It has to be 8, because $-20 + 8$ is -12.

T: Positive 8 or negative 8?

	S: Positive 8. If it were -8, we'd get -28. T: For the distributive property to keep working with negative numbers, a negative times a negative has to give a positive product.
3. Solve multiplication problems with positive and negative rational numbers and summarize the rules for the sign of the product.	
Evaluate each expression. Then summarize the rules for the sign of the product when multiplying positive and negative numbers. a. $6 \times (-7)$ b. $-8 \times (-5)$ c. $-20 \times (1/10)$ <i>Answer: a) -42; b) 40; c) -2; A positive times a positive is positive. A positive times a negative or a negative times a positive is negative. A negative times a negative is positive.</i>	T: Solve each one on your own. Then tell me the rules for the sign of the product. S: (Work.) a is -42, b is 40, c is -2. A positive times a negative is negative, and a negative times a negative is positive. A positive times a positive is positive.